India's air, 2017–2023

What seven years of CPCB monitoring shows about the trend, the season, the day — and whether tomorrow can be forecast

Data science case study | 254,138 city-days | 248 cities | CPCB AQI Repository

Summary

Indian air quality improved between 2017 and 2023, but far less than the seasonal swing that dominates it. On a constant panel of 12 cities, mean AQI fell 17.9% — while November runs 2.56× the monsoon trough within any single year. Nearly one city-day in six is still Poor or worse.

254,138
CITY-DAYS ANALYSED
collapsed from 501,972 station-days across 248 cities

17.9%
AQI FALL 2017-23
on a constant panel of 12 cities

2.6x
SEASONAL SWING
November against the July monsoon trough

16.8%
DAYS POOR OR WORSE
only 14.6% of city-days are Good

Four findings

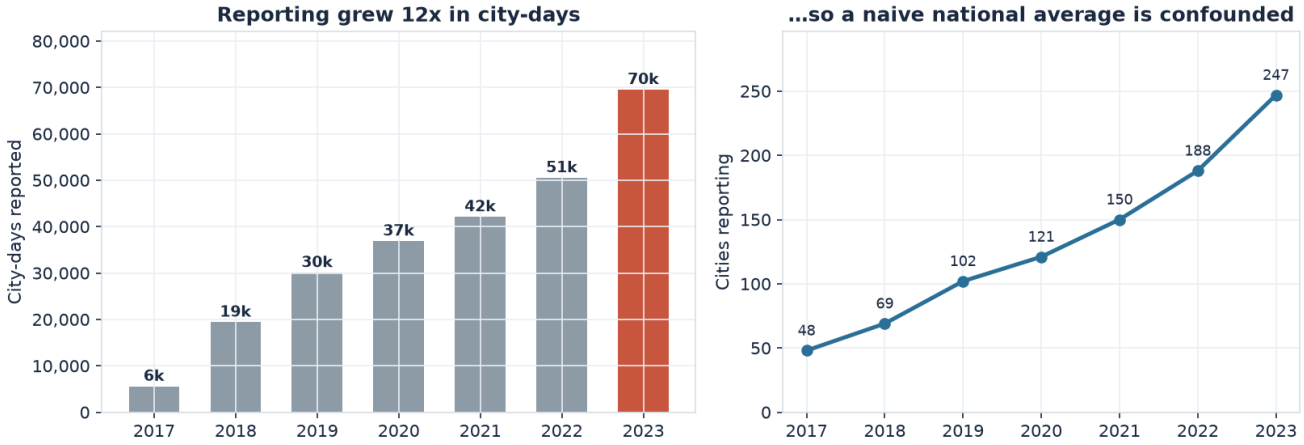
- **The improvement is real but modest.** Mean AQI fell 17.9% on a 12-city constant panel and 12.6% on a denser 52-city panel over 2019–2023. Two independent panels, same direction.
- **Season dwarfs trend.** A 2.6× annual cycle against an 17.9% seven-year movement means a single-month comparison can support almost any conclusion.
- **The daily cycle is atmospheric, not just traffic.** AQI is cleanest at 17:00 and dirtiest at 22:00 — a 41% swing driven by boundary-layer mixing, not rush hour.
- **Forecasting beats persistence by only 8.8%.** A ridge model on lag features wins; gradient boosting adds nothing, which says the remaining error is missing information, not missing model capacity.

The network grew twelve-fold, which fakes a trend

The source is the CPCB AQI Repository: 562,878 station-days of 24-hourly AQI readings for 2017–2023. Cleaning keeps 89.2% of them — a station-day needs at least 18 of its 24 hourly values, and impossible AQI values are dropped. Monitors are then collapsed to one row per city-day using the median, because cities carry up to 39 monitors each and comparing raw rows would simply compare network density.

The decisive problem is coverage. City-days reported rise from 5,575 in 2017 to 69,533 in 2023 — a 12.5× increase as the network expanded from 48 to 247 cities. A national average over all available rows therefore measures who was measuring, not what the air was doing. Every trend in this report uses a constant panel instead.

Figure 1 — The network expanded, which fakes a trend

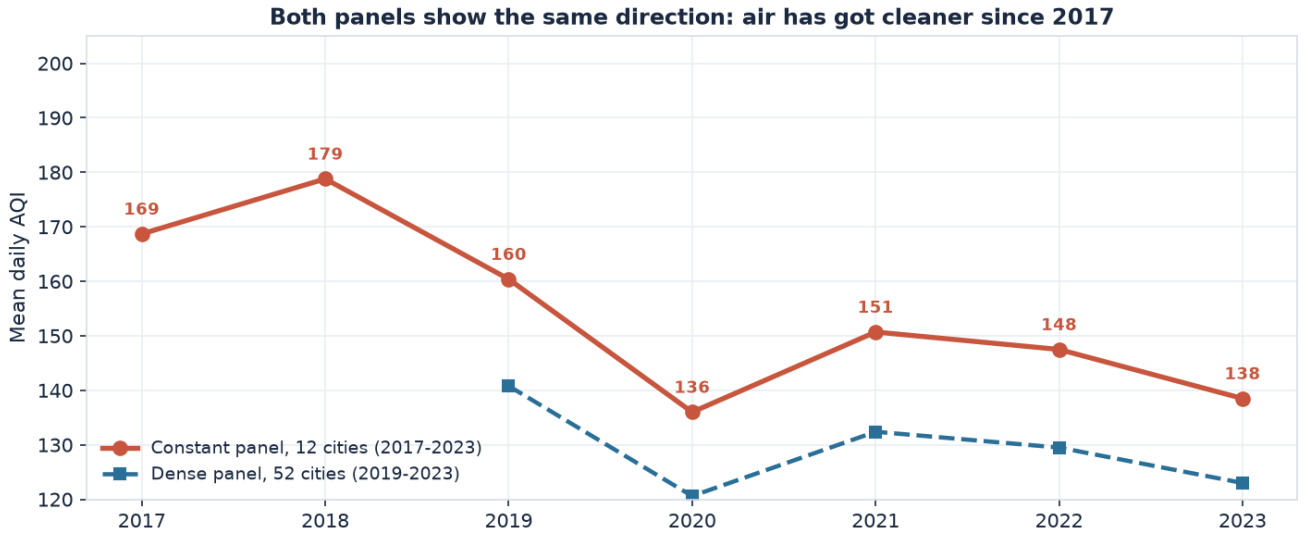


Source: CPCB AQI Repository, 2017–2023; city-days after completeness filtering.

Air quality improved, on both constant panels

On 12 cities with at least 180 reporting days in every year, mean AQI falls from 169 in 2017 to 138 in 2023. The path is not monotonic: 2018 was worse than 2017, 2020 collapsed under lockdown, and 2021–22 rebounded before falling again. A second panel of 52 cities with dense data from 2019 confirms the direction at 12.6%.

Figure 2 — National AQI trend on constant panels



Constant panel: 180+ reporting days in every year 2017-2023. Dense panel: 300+ days, 2019-2023.

9 of the 12 panel cities improved, with a median fall of 13.9%. Varanasi's 65.4% fall is the largest single movement in the data — and should be treated with suspicion, since a monitor relocating or being recalibrated produces exactly that signature. Mumbai and Bengaluru moved the other way.

City	AQI 2017-18	AQI 2022-23	Change
Varanasi	237	82	-65.4%
Jodhpur	219	144	-34.1%
Ghaziabad	270	200	-26.0%
Visakhapatnam	118	122	+2.8%

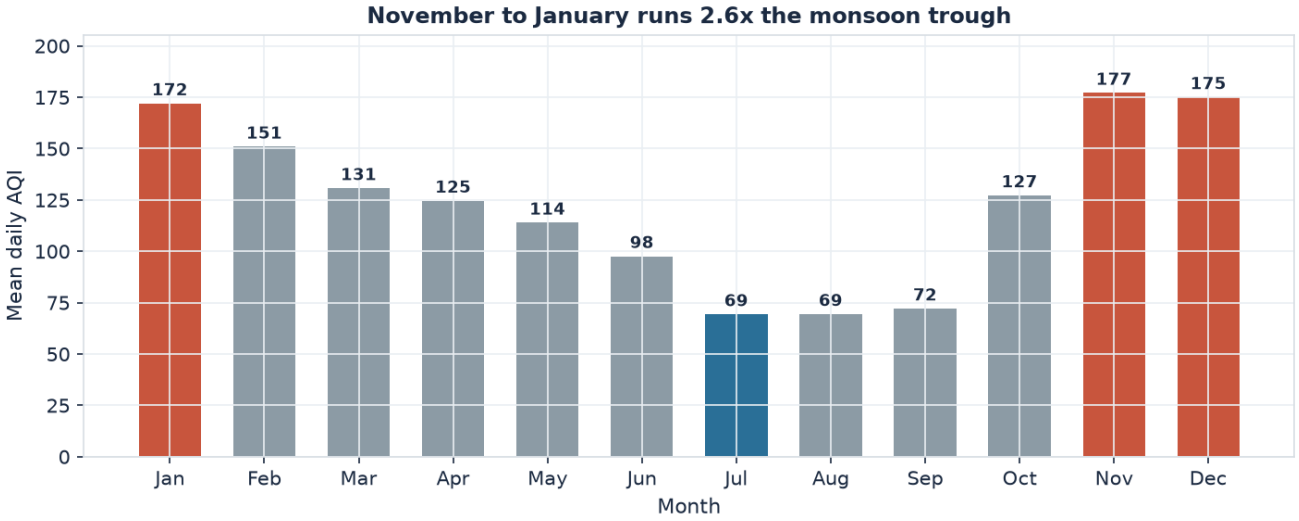
City	AQI 2017-18	AQI 2022-23	Change
Bengaluru	70	73	+4.0%
Mumbai	101	116	+15.1%

Three largest improvements and three largest deteriorations among the constant panel.

Season matters more than the seven-year trend

AQI runs 177 in November and 69 in July — a 2.56× swing inside one year, against an 17.9% movement across seven. Winter pollution in the Indo-Gangetic belt combines a shallow boundary layer, temperature inversions, calm winds and crop-residue burning; the monsoon washes the atmosphere out.

Figure 3 — The seasonal cycle is larger than any seven-year trend

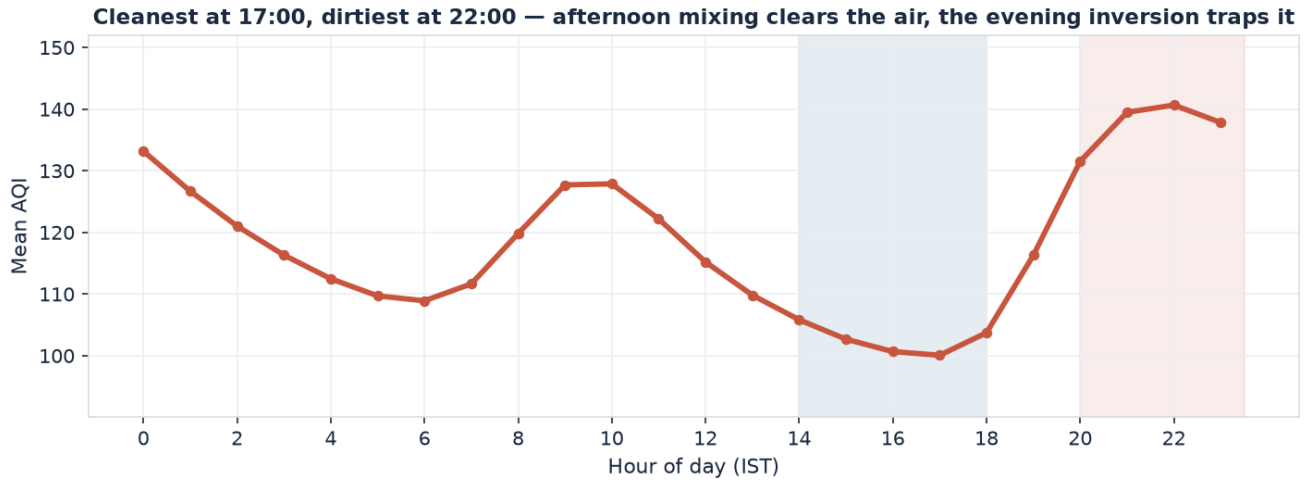


Mean daily AQI by month, all city-days 2017-2023.

Within a day, the air swings 41% — and not at rush hour

AQI bottoms out at 17:00 (100) and peaks at 22:00 (141). The mechanism is the boundary layer: through the afternoon the sun deepens the mixed layer and pollutants disperse into a larger volume; after sunset a stable layer forms and emissions accumulate in a shallower box. Traffic shapes the evening rise; the inversion explains its size.

Figure 4 — Within a day, AQI swings 41%

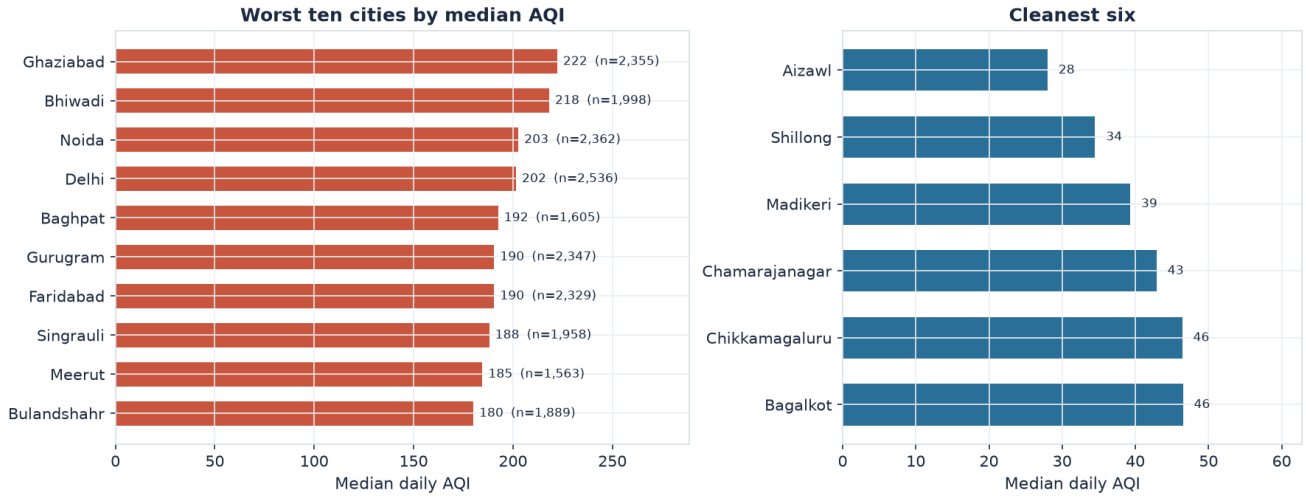


Mean AQI by hour of day, averaged across all 248 cities, 2017-2023.

An eight-fold gap separates the worst city from the cleanest

Among the 115 cities with at least 1,000 city-days, Ghaziabad has a median day of 222 against Aizawl's 28. Set against the CPCB scale, only 14.6% of city-days are Good, while 16.8% are Poor or worse and 5.1% are Very Poor or worse.

Figure 5 — An eight-fold gap between the worst and cleanest cities



Median daily AQI per city; cities with at least 1,000 city-days.

Figure 6 — Where seven years of Indian air actually sits on the CPCB scale

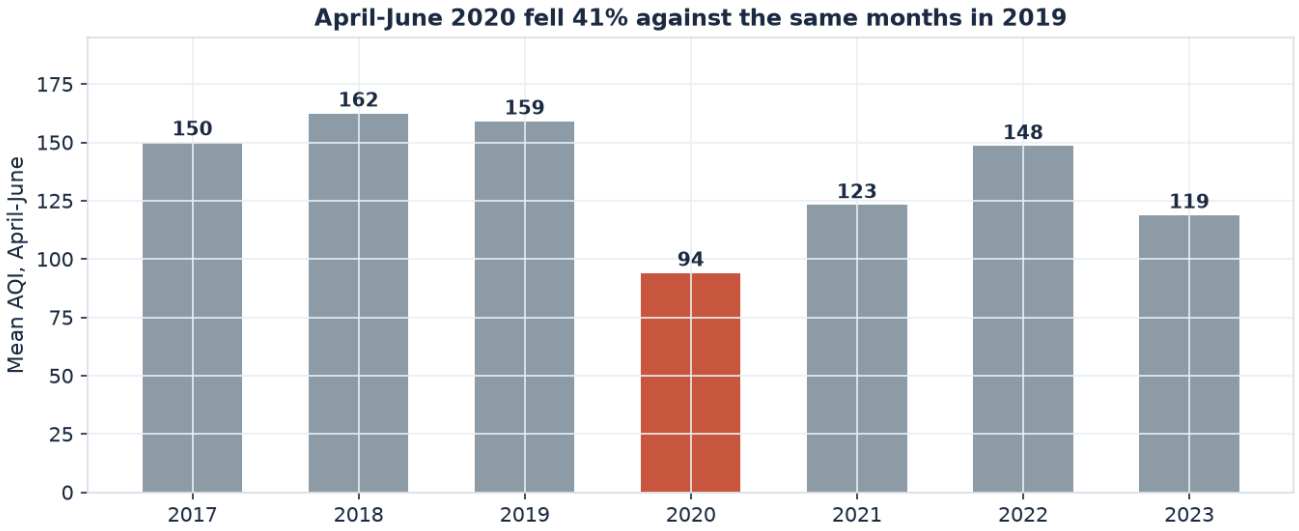


CPCB categories: Good 0-50, Satisfactory 51-100, Moderate 101-200, Poor 201-300, Very Poor 301-400, Severe 401+.

The lockdown removed 41% of the problem in three months

April–June 2020 gave India an unplanned experiment: traffic collapsed, construction stopped, industry idled. On the constant panel, AQI in those months fell to 94 against 159 in the same months of 2019 — a 40.9% drop. The rebound to 148 by 2022 shows how quickly it returns.

Figure 7 — The lockdown experiment



Mean AQI for April-June on the constant 12-city panel.

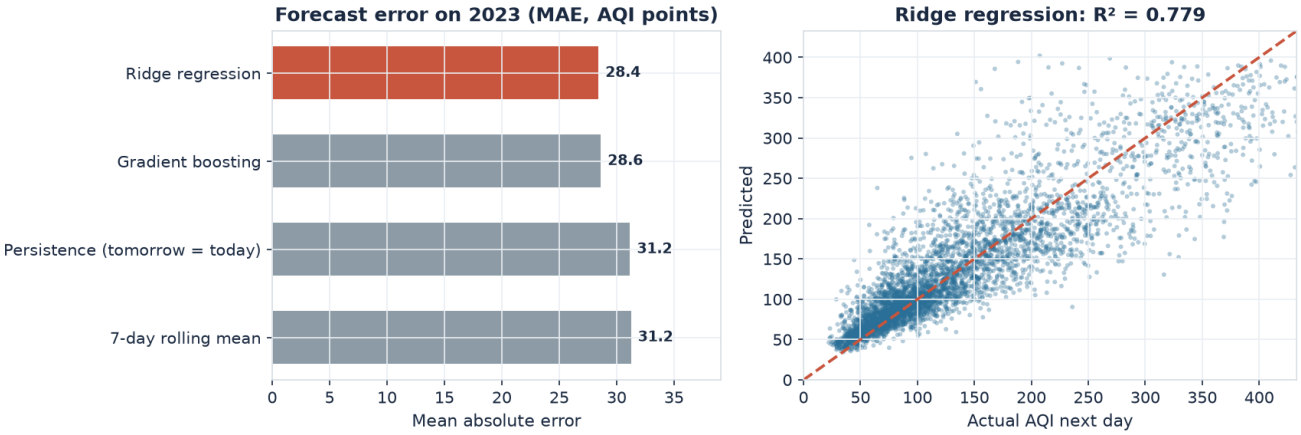
Tomorrow's AQI is predictable, but mostly because today's is

The honest baseline is persistence — tomorrow will be like today. Across 12 pooled cities, trained on everything before 2023 and tested on 2023, a ridge model on lag features reaches MAE 28.4 against 31.2 for persistence — a 8.8% improvement. Gradient boosting, given the same features, does no better (28.6).

Model	MAE	RMSE	R-squared	Within 50
Persistence (tomorrow = today)	31.2	47.4	0.710	80%
7-day rolling mean	31.2	45.2	0.737	80%
Ridge regression	28.4	41.4	0.779	83%
Gradient boosting	28.6	42.4	0.769	82%

Time-based split: train 23,080 city-days to end-2022, test 4,365 city-days in 2023. AQI points throughout.

Figure 8 — Tomorrow's AQI is predictable, but mostly because today's is



Features: AQI lags 1/2/3/7 days, 7- and 30-day rolling means, month, weekday, and a smooth day-of-year term.

Why the ceiling is low

The model has no information the previous month did not already contain. Wind speed, boundary-layer height, rainfall and satellite fire counts explain much of what is left, and none of them are in this dataset. The lesson for anyone building an air-quality app is that a well-specified linear model on lag features is the right level of complexity — additional capacity buys nothing until the feature set changes.

What to do with this

- **Never quote a national average without stating the panel.** The network grew twelve-fold; an unqualified average measures coverage, not air.
- **Compare like with like, and compare within a season.** A November-to-November comparison is the only one that isolates policy from meteorology.
- **Time outdoor activity to the afternoon trough.** The 17:00 minimum is a 41% improvement over 22:00 — larger than most policy interventions achieve in a year.
- **Spend the next effort on data, not models.** Adding meteorology and fire counts is worth more than any model change tested here.

Limitations

- **AQI is an index, not a measurement.** It reports the worst sub-index across pollutants, so it hides which pollutant drove a given day; this dataset carries no pollutant breakdown.
- **One number per city per day.** A median across monitors erases the intra-urban variation that actually determines personal exposure.
- **Station siting changes.** A relocated or recalibrated monitor creates a step change that looks like a trend — Varanasi's fall should be read that way until confirmed.
- **2020 is a structural break.** Models trained across it learn a level that never recurs.
- **No meteorology.** Without wind, mixing height and rainfall the forecast ceiling stays low.

Next steps

- Join pollutant-level data (PM2.5, PM10, NO2) to identify what drives the index.
- Add meteorology and satellite fire counts — the features most likely to beat persistence by more than 9%.

- Model at station level with a hierarchical structure rather than collapsing to city medians.
- Test a 3-day horizon, where persistence decays and a real model earns its keep.

Reproducing this analysis

Everything here is generated from one notebook. It reads the raw CPCB parquet, streams it in batches to stay inside memory, applies the cleaning rules, builds all eight figures, trains and evaluates the four forecast candidates, and writes the metrics file this report reads from — so no number here was typed by hand.

File	Contents
india_air_quality_analysis.ipynb	Full analysis, executed, with outputs
data/cpcb-aqi.parquet	Raw CPCB station-day data, 562,878 rows
metrics_air.json	Every figure quoted in this report
figures/	Eight charts, 150 dpi

Runtime is about one minute on a single CPU. Random seed 42; scikit-learn 1.9, pandas 3.1. Source: CPCB AQI Repository, mirrored at github.com/Vonter/india-cpcb-aqi.